

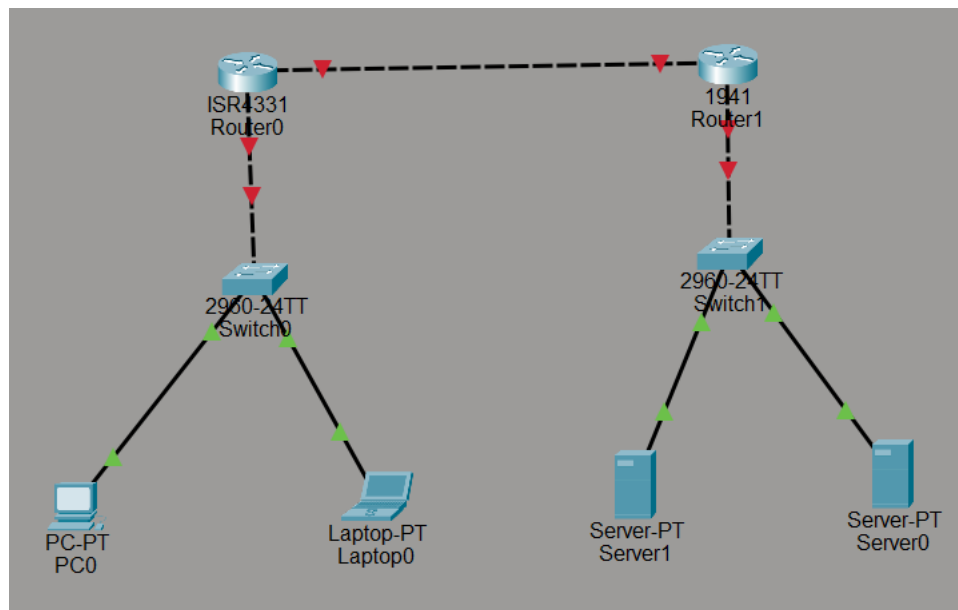
TUTORIAL NO. 4: DESIGNING A NETWORK MANAGEMENT TOOL FOR A GNS3 SIMULATED NETWORK.

Prerequisite:

- Installation of Oracle Virtual box
- Installation of GNS3

Part 1: Network node creation and interconnection

Consider the network architecture of an organization below:



Network Configurations

Subnet the network address 192.168.1.0/24 to create four subnetworks. One subnet will be for:

- the link between Router 1 and Router 0
- VLAN 10 containing the servers (The DMZ zone)
- VLAN 20 containing the laptop (The internal network)
- VLAN 30 containing the PC as our NMS server (Our NOC department)

Configure static route(s) between Router 0 and Router 1 (if necessary).

Create an Inter-VLAN between the VLANs 10, 20 and 30 to ensure communication is possible across all devices in the network.

Perform the following SNMP Configurations where necessary:

- ***snmp-server community public RO*** # to configure the public community string for read-only
- ***snmp-server community private RW*** # to configure the public community string for read-only

- ***snmp-server host <server_ip> version 2c public** # directs the device where to send the traps (i.e. to monitoring server.*
- ***snmp-server enable traps** # enables the device to send trap messages to the monitoring server*

Part 2: Collecting network events via SNMP from network devices for analysis and Storage in a DB (Backend)

- 1) We need to create a virtual machine in Oracle Virtual box,
- 2) Install Windows on the VM. This will act as our NMS (dashboard + DB)
- 3) To allow the VM to connect to the network, a new interface needs to be added in the Network Panel of the VM. There, we disable the NAT, create a second network interface and select Virtual Box Host Only Adapter.
- 4) Using any programming language(s) of your choice, write a code to collect events from the different network devices periodically, analyze those that are important and store this information in a database. This will be the backend of your application.

Part 3: Collecting stored data (events) from the DB and displaying it on a Dashboard to the network administrator (Frontend)

- 1) Retrieve these events/notifications periodically and update them on a dashboard for the network administrator to be informed as soon as possible of the current faults in the network (Fault management)
- 2) Still on the dashboard, provide a tab for inventory of equipments/nodes in the network (Audit management).
- 3) Calculate the average traffic on a network link generated by a given IP packet, as create a billing system for each IP address (Service level agreement and billing).
- 4) Plus+ : Can you create a small tab, that will permit “auto-discovery” as we saw with PRTG monitoring tool (i.e. to add new devices at will) ?

Compare your project in terms of customization (added value) and performance to leading market tools like Solarwinds, Nagios, PRTG, Cacti etc.